Local Systems over Graphs and CM cycles over Shimura Curves

Workshop on Arakelov and Diophantine Geometry

August 4, 2026 Brown University

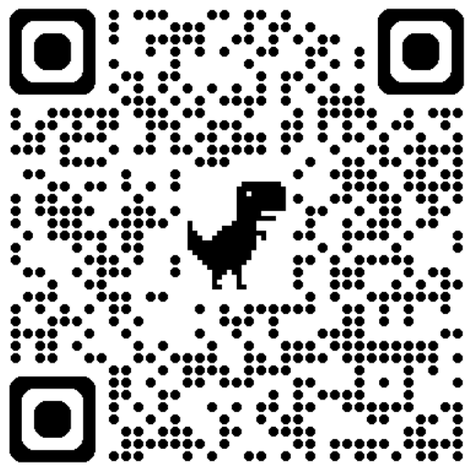
Boya Wen (University of Richmond)

Two Updates !

if you were at my talk(s) of a similar title ...

1. Draft available.

   Comments Welcome !

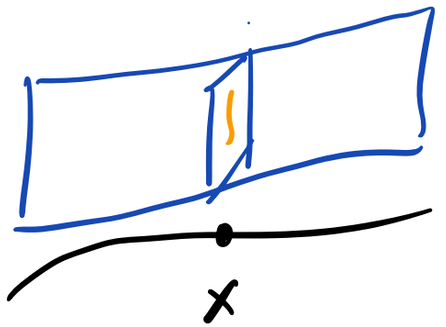


If not, the talk is for you, too !

2. Concrete dictionary

Kuga Variety over Shimura Curves	Local System on Graphs
⋮	⋮

Why Kuga Varieties over Shimura curves ?



The Gross - Zagier - Zhang Formula

Height pairings of Heegner divisors on modular curves

Heegner cycles in Kuga-Sato varieties over modular curves

CM cycles in Kuga varieties over some Shimura curves

Derivatives of certain L-series associated to

(with reference to an imaginary quadratic field K)

Certain weight -2 modular forms.

$2k$

Level $\Gamma_0(N)$

Heegner condition : $p \mid N \Rightarrow p$ splits in K

Generalize to $N = N_- N_+$?

(N_-: even number of distinct primes, inert in K; N_+: split in K)

($k=1$ covered in Yuan-Zhang-Zhang)

More details of the geometric objects

Let B be a quaternion algebra over $\mathbb{Q}$ unramified at ∞, ramified at $p \mid N_-$ (even number of distinct finite primes).

X_{N_-, N_+}: Moduli space for abelian surfaces with

- $\iota: \mathcal{O}_B \hookrightarrow \mathrm{End}(A)$
- principal polarization
- N_+ level structure

CM points by $\mathcal{O}_K$:

$\mathcal{O}_K \hookrightarrow \mathrm{End}_{\mathcal{O}_B}(A)$

Let X denote a finite cover of X_{N_-, N_+} over which there exists an universal abelian surface $\mathcal{A} =: Y$. Let F be a number field s.t. $\mathcal{O}_K$-CM pts are rational and X and Y have regular integral models $\mathcal{X}$ and $\mathcal{Y}$ over $\mathrm{Spec}\, \mathcal{O}_F$.

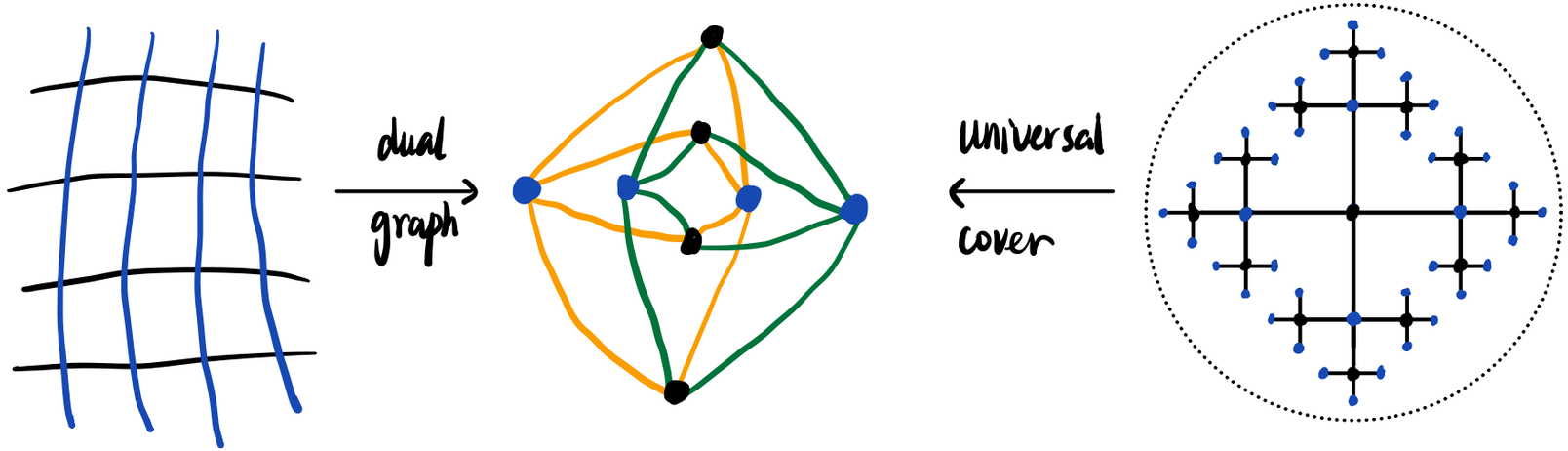
Connection to graphs: (Cerednik–Drinfeld)

At a prime $\mathfrak{P}$ of $\mathcal{O}_F$ above a prime $p \mid N_-$, connected components of the reduction of the Shimura curve are collections of $\mathbb{P}^1$'s intersecting transversally, in an arrangement whose dual graph is a finite regular graph of degree $p+1$.

$\mathbb{P}^1 \longleftrightarrow$ vertex

intersection $\longleftrightarrow$ edge

An example where $p=3$ is a Drinfeld prime



Now Row 1 of the Dictionary!

Kuga Variety over Shimura Curves	Local System on Graphs
The connected component of $X_{\bar{\mathbb{F}}}$ that $\bar{x}$ hits, for $\mathbb{F}$ above a prime $p \mid N$.	A regular graph $G = \Gamma \backslash \mathcal{T}$ of degree $p+1$
⋮	⋮

dual graph

$\mathbb{F}'$ components $\longleftrightarrow$ vertices

intersections $\longleftrightarrow$ edges

CM cycles

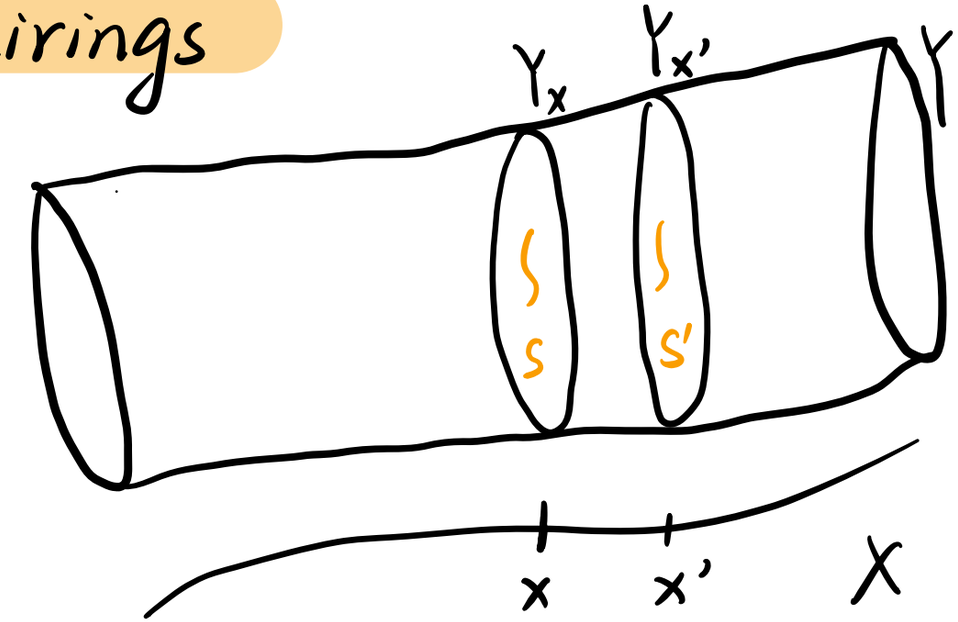
Given an abelian surface A with $L : B \hookrightarrow \mathrm{End}^\circ(A)$ (= $\mathrm{End}(A) \otimes \mathbb{Q}$) and principal polarization given by the ample divisor D. Let $NS(A)_{\mathbb{Q}}^{ord}$ (3-dim) denote the subspace of $NS(A)_{\mathbb{Q}} := NS(A) \otimes \mathbb{Q}$ spanned by $\{[\beta^* D] \mid \beta \in B^{\times}_{tr=0}\}$.

Let $NS(A)_{\mathbb{Q}}^{CM}$ denote the orthogonal complement of $NS(A)_{\mathbb{Q}}^{ord}$ in $NS(A)_{\mathbb{Q}}$.

$$\dim NS(A)_{\mathbb{Q}}^{CM} = \begin{cases} 0 & \text{if } \mathrm{End}_B^\circ(A) = \mathbb{Q} \\ 1 & \text{if } \mathrm{End}_B^\circ(A) = K \quad (\text{CM, char } 0) \\ 3 & \text{if } \mathrm{End}_B^\circ(A) = B' \quad (\text{supersingular, char } p) \end{cases}$$

our CM cycle is here ↑

Height Pairings



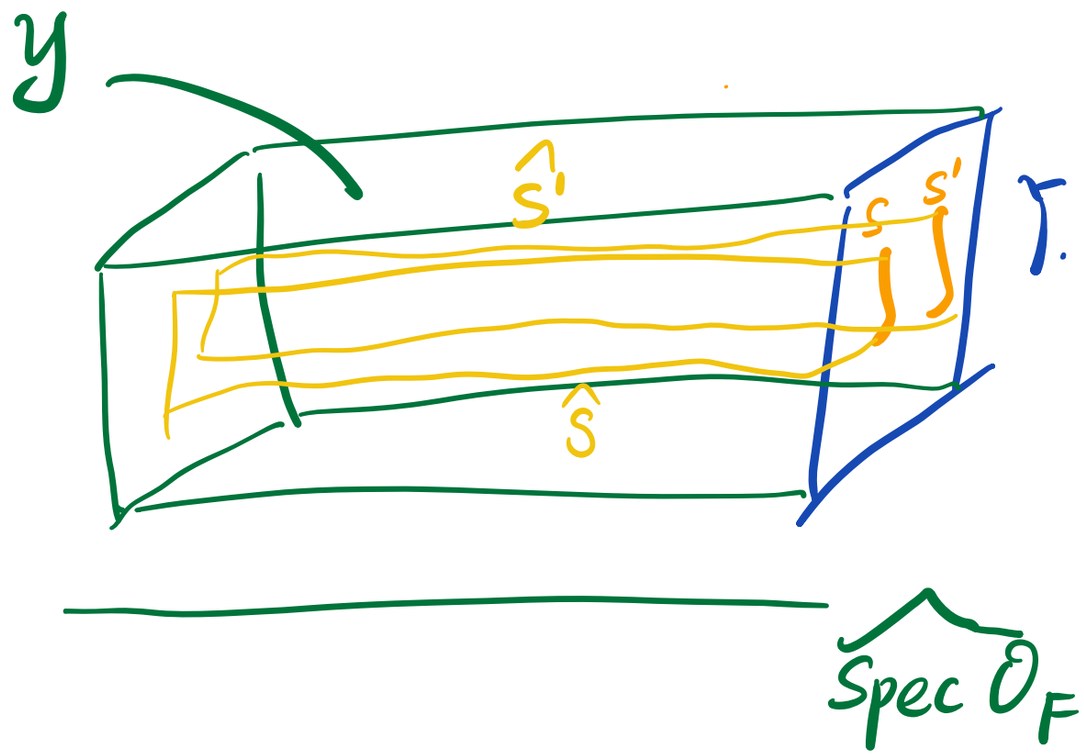
$\operatorname{codim} S + \operatorname{codim} S'$

$= \dim Y + 1$

Arakelov

Faltings

Gillet–Soulé

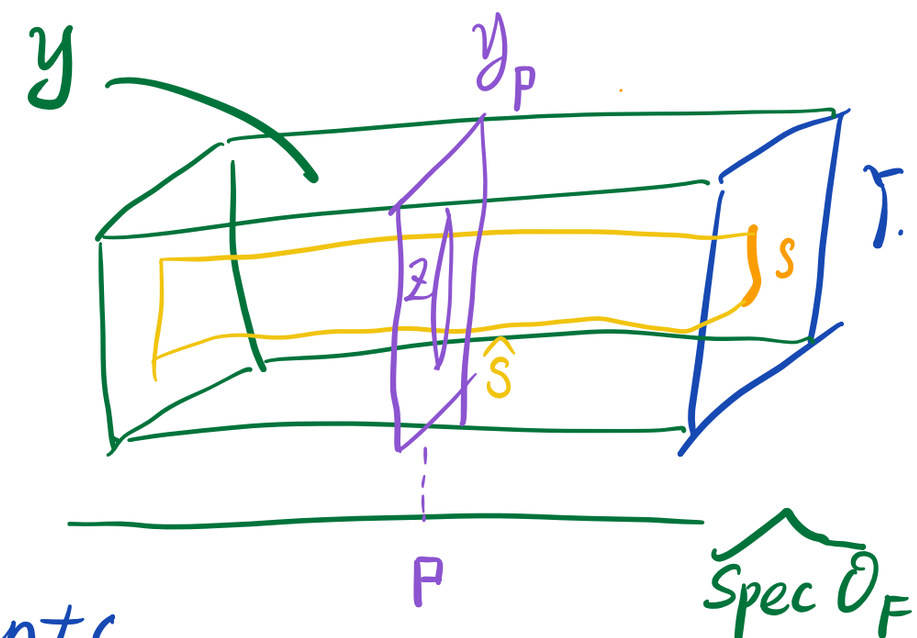


Choosing appropriate arithmetic extensions

Before we can simply define $\langle s, s' \rangle$ to be $\langle \hat{s}, \hat{s}' \rangle$

Need: $\langle \hat{s}, \hat{s}' \rangle = \langle \hat{s}, \tilde{s}' \rangle$ if $\hat{s}'$, $\tilde{s}'$ are both arithmetic extensions of s'.

Find $\hat{s}$ s.t. $\forall$ any cycle Z supported in special fibers, $\langle \hat{s}, Z \rangle = 0$.



Unfortunately, $\bar{s}$ doesn't work. Fortunately, only fails at $F|N_-$, where we need to find adjustments.

More Puzzle Pieces

Kuga Variety over Shimura Curves	Local System on Graphs
Finding adjustment cycles R_p s.t. $\hat{S} = \bar{S} - \sum_{p \mid N} R_p$ satisfies $(\hat{S} \cdot Z)_p = 0$ for any Z supported on special fibers	?
Given two CM cycles S, S', the local intersection number of an adjustment cycle of S with $\bar{S}'$	?

More Puzzle Pieces

Kuga Variety over Shimura Curves	Local System on Graphs
The connected component of $X_{\square}$ that $\bar{x}$ hits, for $\mathfrak{P}$ above a prime $p \mid N$.	A regular graph $G = \Gamma \backslash \mathcal{T}$ of degree $p+1$
?	A local system of spheres $\mathcal{S}$ above G.
?	The local system of $\mathcal{H}_1(\mathcal{S})$ (deg-1 spherical harmonics)

dual graph

$\mathfrak{P}'$ components $\longleftrightarrow$ vertices

intersections $\longleftrightarrow$ edges

What is a local system over graph ?

Attach to each vertex v,

an inner product space $\mathcal{L}(v)$,

to each edge e,

an inner product space $\mathcal{L}(e)$,

whenever v is an end of e,

an isometric isomorphism $\varphi_{ev} : \mathcal{L}(e) \to \mathcal{L}(v)$

We do this over $\mathbb{R}$ (or $\mathbb{C}$) — how to achieve this at the finite places? By building $\mathbb{Q}$ v.s. !

Local system of spheres over a graph.

Attach to each vertex v, an oriented real 2-sphere S_v,

to each edge e, an oriented real 2-sphere S_e,

whenever v is an end of e,

a bijective orientation-preserving isometry $\varphi_{ev}: S_e \to S_v$.



Induces a local system $L^2(S)$ of real-valued square-integrable functions on the spheres, which decomposes into irreducible $SO(3)$ representations $H_n(S)$ ← $2n+1$ dim'l.

↑

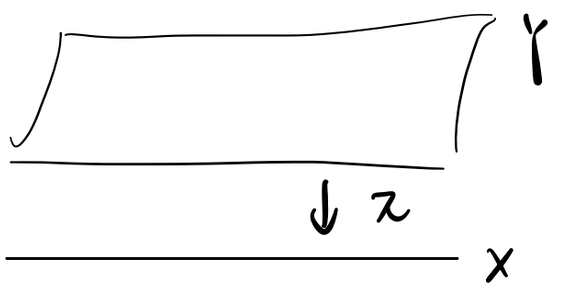
Homogeneous harmonic polynomials of degree n in $\mathbb{R}[x,y,z]$.

vanishing under $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

Eg. $H_1(S^2) = \{ax + by + cz \mid a, b, c \in \mathbb{R}\}$

Connection to Shimura Curves.

Suppose $K = \mathbb{Q}(\sqrt{-d})$.



reduction

mod $\mathfrak{p}$

(above $p \mid N_-$)

(Cerednik - Drinfeld
k. Li)

$$\pi^{-1}(C_i) \sim C_i \times E_i^2$$

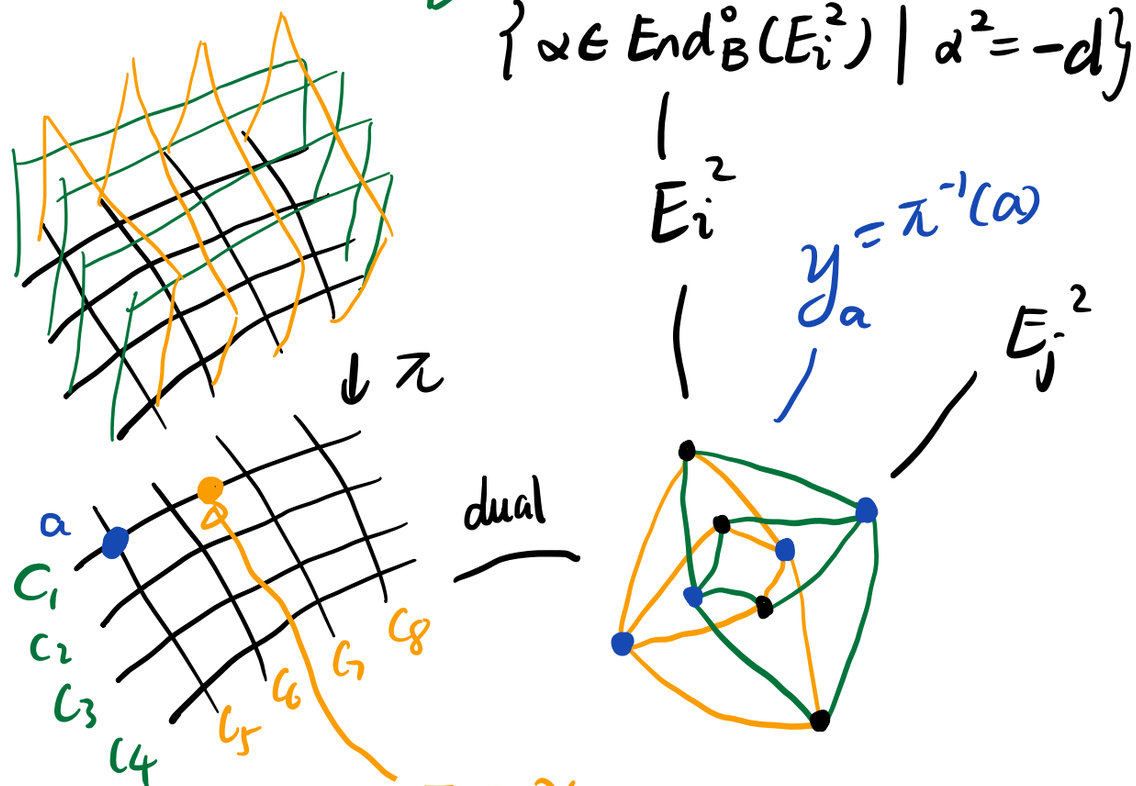
$\mathfrak{P}$ s.s.e.c.

$H_1(S)$ → $NS(E_i^2)$ $\otimes_{\mathbb{Z}}^{CM}$ $\mathbb{R}$

$\bar{S} \cap y_{\mathfrak{p}}$ push forward here.

Sphere →

$\{\alpha \in \mathrm{End}^0_B(E_i^2) \mid \alpha^2 = -d\}$



$\bar{x} \cap \mathcal{X}_{\mathfrak{p}}$ is not at the crossings (Molina, 2012)

Summary :

Kuga Variety over Shimura Curves	Local System on Graphs
The connected component of $\chi_{\mathfrak{P}}$ that $\bar{x}$ hits, for $\mathfrak{P}$ above a prime $p \mid N$.	A regular graph $G_J = \Gamma \backslash \mathcal{T}$ of degree $p+1$
The system of CM structures above the said connected component	A local system of spheres S above G_J.
The local system of $NS_{\mathbb{R}}^{CM}$	The local system of $\mathcal{H}_1(S)$ ↑ deg-1 spherical harmonics

dual graph

$\mathfrak{P}'$ components $\longleftrightarrow$ vertices

intersections $\longleftrightarrow$ edges

W. 26+

Recall our remaining Goals:

Kuga Variety over Shimura Curves	Local System on Graphs
Finding adjustment cycles R_p s.t. $\hat{S} = \bar{S} - \sum_{p \mid N} R_p$ satisfies $(\hat{S} \cdot Z)_p = 0$ for any Z supported on special fibers	?
Given two CM cycles S, S', the local intersection number of an adjustment cycle of S with $\bar{S}'$	?

Finding R_P

Want: $(\bar{S} \cdot Z)_P = (R_P \cdot Z)_P \quad \forall Z$ of appropriate dimension (2) supported on y_P

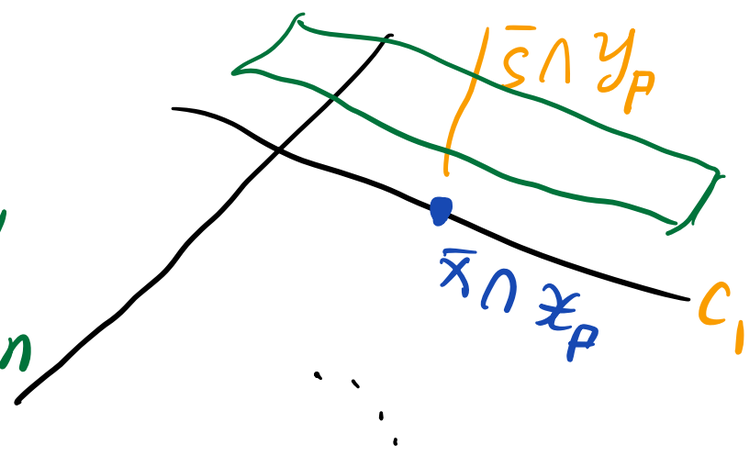
$$R_P = \sum_{i=1}^{n} f_i^* (C_i, R_i)$$

Find them

$f_i : \pi^{-1}(C_i) \to C_i \times E_i^2$

The only cycles w/ nonzero intersection w/ $\bar{S}$ is of the form $f_1^*(C_1, T)$

$\bar{x}$ Φ hits



When computing $(f_1^*(C_1, T) \cdot f_1^*(C_1, R_1))_P$, need to move to neighboring $\mathbb{P}^1$ components

Harmonic analysis on local systems over graphs

Given a local system $\mathcal{L}$ over a graph whose set of vertices is V, define $\Delta : \Gamma(V, \mathcal{L}) \to \Gamma(V, \mathcal{L})$ by

$$\Delta f(v) = \sum_{\substack{e \text{ having} \\ v, v' \text{ as ends}}} f(v) - \varphi_{ev}\varphi_{ev'}^{-1} f(v')$$

We have $\Gamma(V, \mathcal{L}) = \mathrm{Im}(\Delta) \oplus \boxed{\ker(\Delta)}$ (orthogonal; $= 0$ if monodromy representation is irred. & nontrivial)

So for any $\phi \in \mathrm{Im}(\Delta)$, there exists a unique $f \in \mathrm{Im}(\Delta)$ s.t. $\Delta f = \phi$. We can solve such "differential equations" by constructing Green's functions for local systems over graphs

Summary (Continued) (W. 26+)

Kuga Variety over Shimura Curves	Local System on Graphs
Finding adjustment cycles R_p s.t. $\hat{S} = \bar{S} - \sum_{p \mid N} R_p$ satisfies $(\hat{S} \cdot Z)_p = 0$ for any Z supported on special fibers	Solving a vector-valued "differential equation" in the local system (using vector-valued Green's functions constructed in W. 22)
Given two CM cycles S, S', the local intersection number of an adjustment cycle of S with $\bar{S}'$	A multiple of scalar-valued Green's function (constructed in W. 22, need input from sphere)

Thank you!